SESSION REPORT

Session d94c9109

10/13/2026, 8:26:48 AM · 145.16 s · jc4880p433-heart

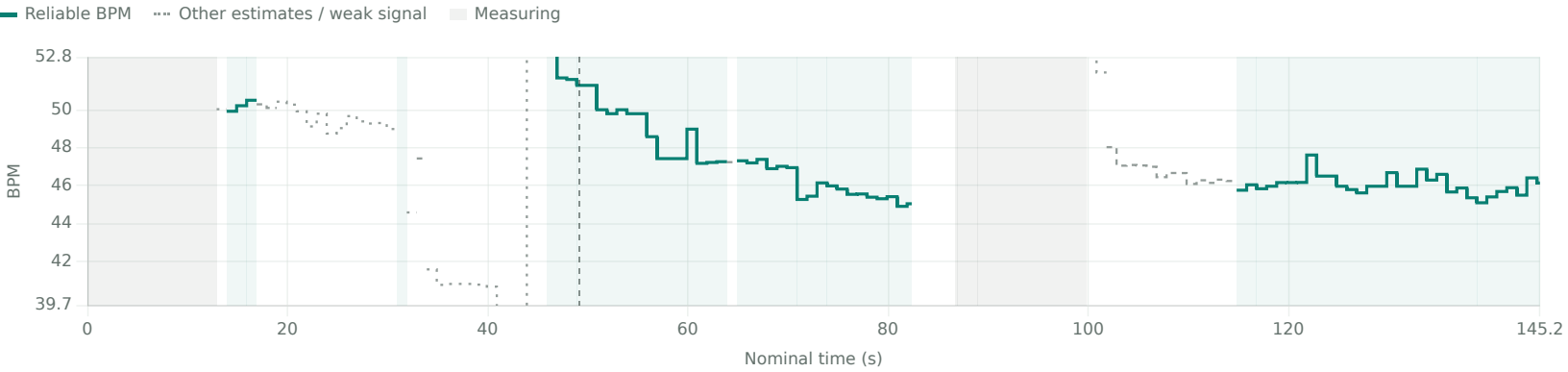
48.1% reliable coverage (69.78 s of 145.16 s). Summary statistics describe reliable periods only. Other estimates remain available for inspection.

Reliable mean	Reliable range	Reliable coverage	First usable reading	Detected peaks
47.2 BPM	**44.9-53.1 BPM**	**48.1 %**	**13.91 s**	**123**
Nominal-duration weighted	Reliable periods only	Not whole-session confidence	From recording start	Not verified heartbeats

Session range

0.0-145.2 s

Heart rate



Nominal time	Recorded BPM	Reliability	Method	Red-channel BPM	Correlation	Period distinctness	Peak dispersion	IR RMS
49.19 s	**51.3**	**Reliable**	**PEAKS**	**50.9**	**0.887**	**0.887**	**0.030**	**22.5 counts**

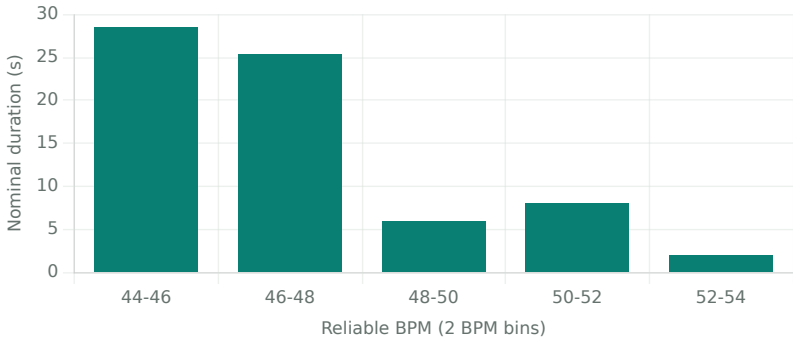
Flags: None · Reading at 48.91 s · Estimator window 38.04-48.04 s

Reliable periods

Recorded policy, not a clinical assessment

Period	Start	End	Duration
1	13.91 s	16.91 s	3.00 s
2	30.91 s	31.91 s	1.00 s
3	45.91 s	63.91 s	18.00 s
4	64.91 s	82.37 s	17.46 s
5	114.84 s	145.16 s	30.32 s

Reading distribution

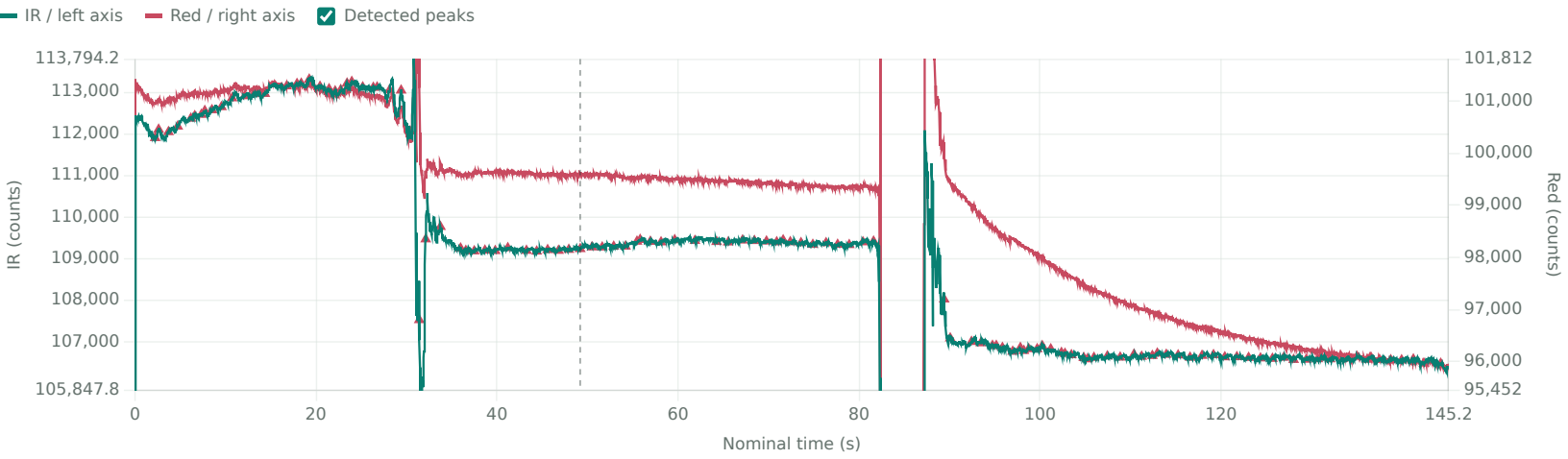


Capture at a glance

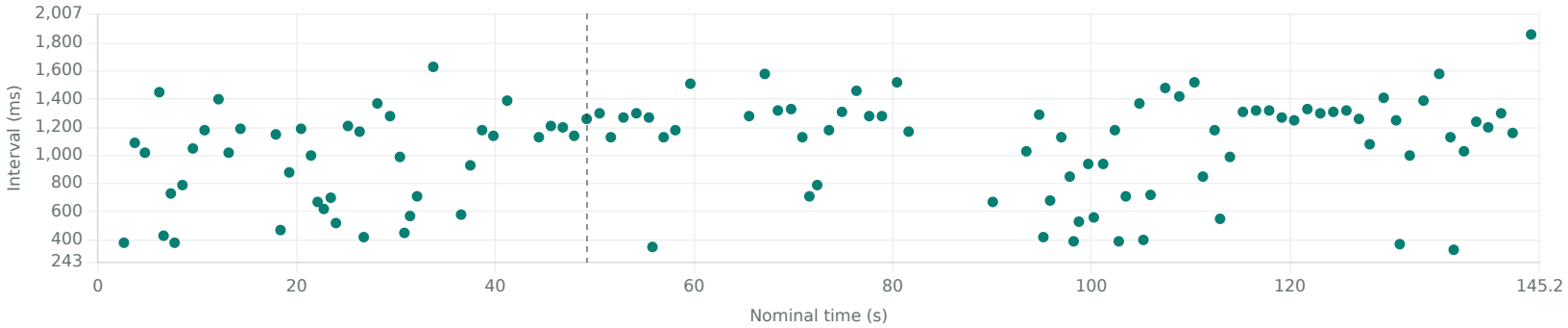
Reliable median	46.4 BPM
Usable signal	86.78 s (59.8%)
Longest reliable period	30.32 s
Stored samples	14,516
Lost samples	0
FIFO overflow counter	0
Capture status	Complete

Optical pulse estimates are not ECG measurements. Detected peaks and in-range intervals are not independently verified heartbeats. No clinical HRV or oxygen-saturation result is inferred.

Optical signals



Detected peak intervals



49.19 s	1260 ms	Within detector bounds	Reliable
Nearest detected peak	Interval	Interval check	BPM at detection

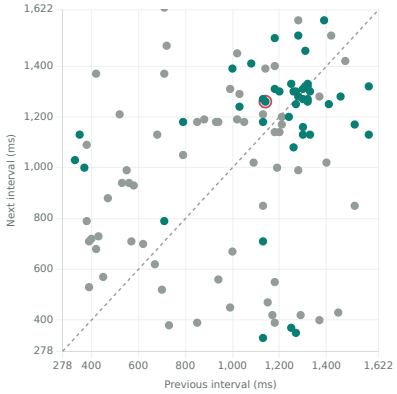
Pulse variability Experimental

Detected optical pulse intervals, not verified ECG heartbeats. Detector bounds and reliable BPM do not establish interval accuracy. Clinical HRV metrics are unavailable for this capture.

122	115	7	108
Interval records	Detector-bounded intervals	Rejected / interrupted	Adjacent pairs

Poincaré plot

Reliable BPM at all three peaks Other detector-bounded pairs

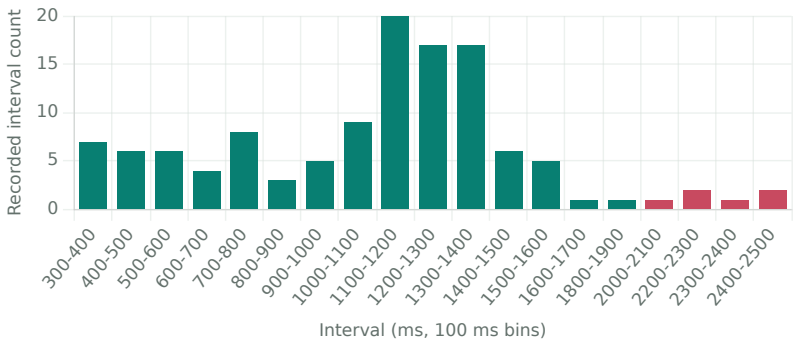


108 of 108 adjacent pairs; 2 outside the visible scale. Equal interval axes; diagonal = unchanged interval. All values restores extremes.

46.79-49.19 s · 1140 -> 1260 ms · Change 120 ms · Reliable BPM at endpoints

Interval distribution

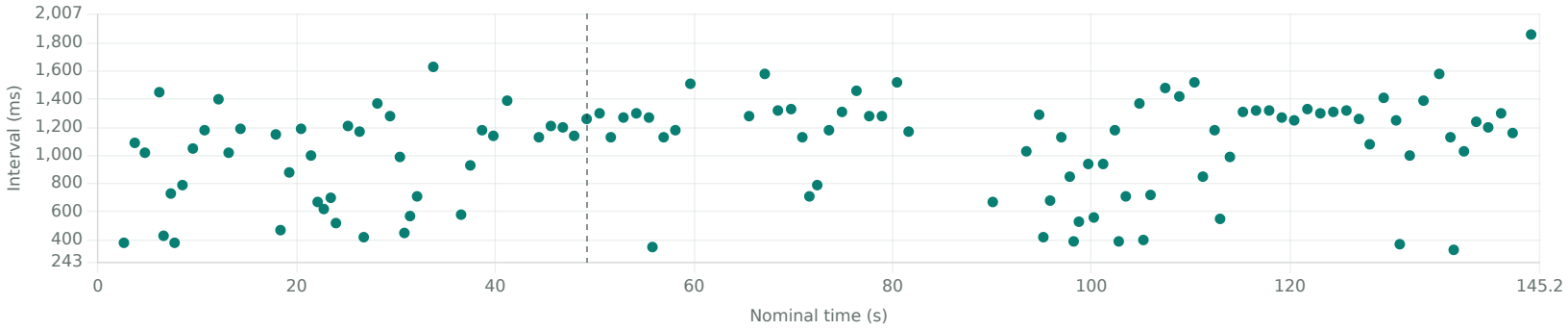
Within detector bounds Rejected / interrupted



Scope	Fully contained intervals in selected range
Timing step	10 ms
Unavailable interval durations	1
Longest detector-bounded run	51.84 s
Pairs with reliable BPM at all endpoints	45
Validated normal intervals	Not established
Artifact correction	None; recorded intervals retained

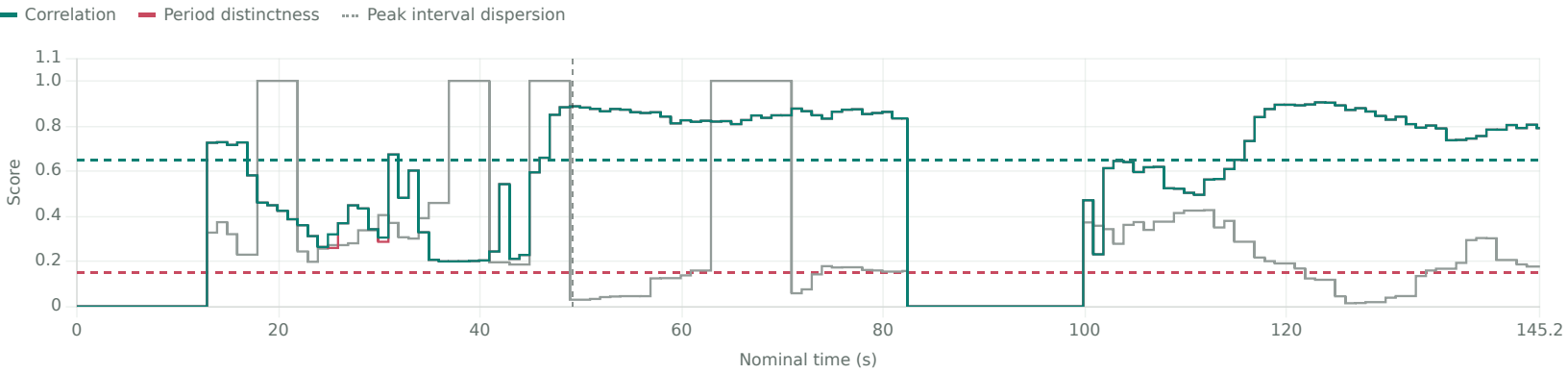
RMSSD, SDNN and pNN50: unavailable. Validated beat intervals and documented artifact handling are required. No stress or recovery score is inferred.

Pulse interval timeline



index time.

Estimator evidence



Session events

Time	Event	Status	Quality flags
	recording start		
	quality change	measuring	settling
	quality change	weak signal	abrupt change
	quality change	measuring	settling
	quality change	measuring	settling
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	Reliable	unstable intervals
	quality change	usable	rate disagreement, unstable intervals
	quality change	usable	unstable intervals
	quality change	weak signal	unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	usable	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals, stale beat
	quality change	weak signal	low periodicity, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	unstable intervals
	quality change	Reliable	unstable intervals
	quality change	Reliable	None
	quality change	Reliable	unstable intervals
	quality change	usable	unstable intervals, stale beat
	quality change	Reliable	unstable intervals
	quality change	Reliable	None
	quality change	Reliable	unstable intervals
	quality change	weak signal	abrupt change
	quality change	weak signal	no contact

Time	Event	Status	Quality flags
	quality change	measuring	settling
	quality change	weak signal	abrupt change
	quality change	measuring	settling
	quality change	weak signal	abrupt change
	quality change	measuring	settling
	quality change	measuring	settling
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	usable	rate disagreement, unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	Reliable	unstable intervals
	quality change	Reliable	None
	quality change	Reliable	unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	Reliable	unstable intervals
	recording stop		

Acquisition

sample rate hz	100
sensor sample rate hz	100
sample average	1
time basis	stored_sample_index_not_hardware_sequence
read elapsed us basis	host_fifo_poll_not_sample_timestamp
sensor	MAX30101
sensor address	87
part id	21
revision	6
pulse width us	411
adc range na	4096
led red code	31
led ir code	31
fifo rollover	false
max poll gap us	12678
fifo overflow counter	0
ambient overflow events	0
lost samples	0
warnings	None

Estimator & reliability policy

version	heart_bpm_v1
window s	10
startup s	2
update s	1
min bpm	30
max bpm	180
highpass hz	0.5
lowpass hz	4
min correlation	0.35
min rms counts	2
refractory s	0.3
min contact counts	5000
waveform	causal_second_order_highpass_then_lowpass_ir_counts
Policy: version	heart_reliable_v1
Policy: requires complete capture	true
Policy: min correlation	0.65
Policy: min period distinctness	0.15
Policy: max red rate relative difference	0.15
Policy: max peak interval dispersion	0.15
Policy: minimum summary duration s	10
Policy: allows autocorrelation with unstable peak intervals	true
Policy: excluded quality flags	settling, no contact, clipped, flatline, abrupt change, low amplitude, low periodicity, ambiguous period, red disagreement, stale beat, out of range

Time basis & integrity

CSV pairing	Verified: sample count, contiguous indices, byte size and CRC32
CRC32	9bc0b2b4
Nominal duration	145.16 s
Host duration	144.625 s
Maximum host polling gap	12.678 ms
Firmware	1.40.0
Stop reason	manual stop
Exercise stop marker	Not recorded
Annotations	[]

Time uses stored sample index / effective sample rate. FIFO read elapsed time is host polling time, not a hardware sample timestamp. The report does not repair or re-estimate the capture.